CN5303 Industrial Case Studies for Renewable Energy Systems

A Case Study of Low-Carbon Data Centre Deployment in Singapore

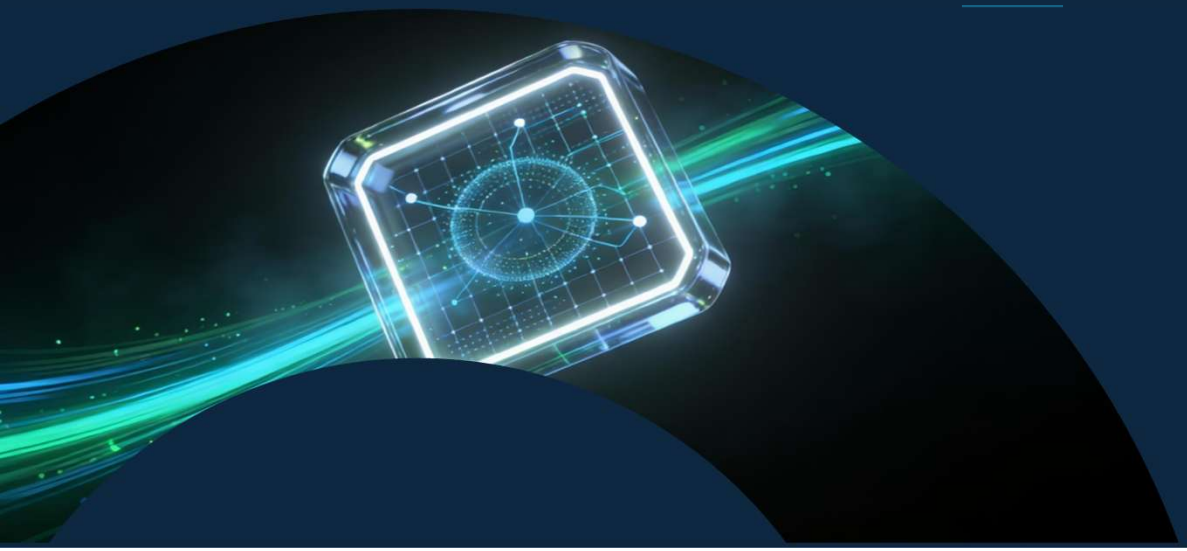


Group 5

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CONTENTS



01 INTRO

02 DECARBONIZATION OPTIONS

03 STRATEGIC CHALLENGES

04 INTEGRATED ASSESSMENT

05 CONCLUSION

Singapore's Data Center Landscape

Strategic Importance to Singapore

Enabling Economy-Wide Digitalization

Provides essential computing infrastructure for digital activities across industries.

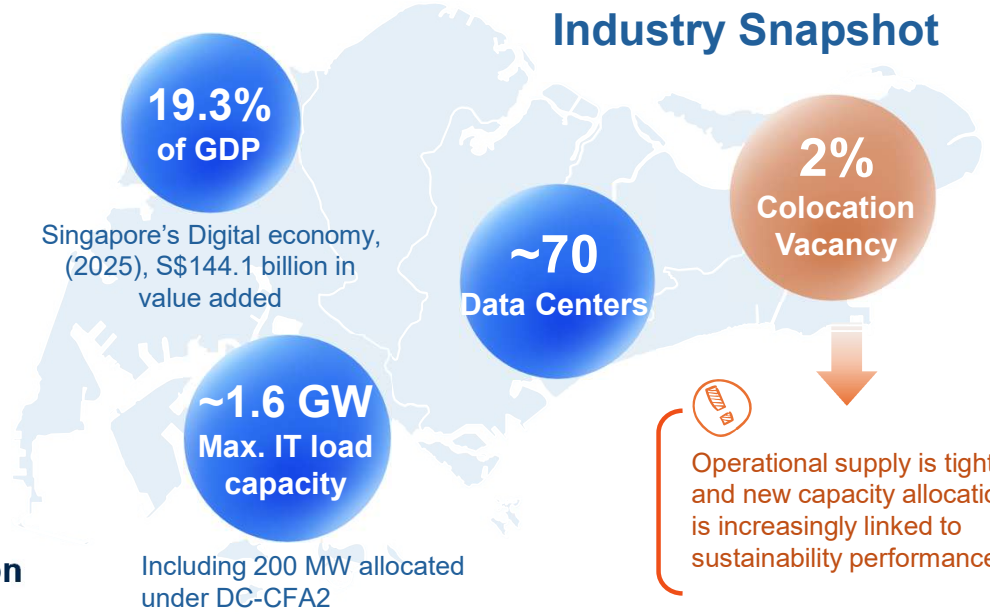
Supporting Critical Domestic Workloads

Meets latency, secure and resilient needs in sectors such as finance, healthcare and AI.

Strengthening Singapore's Regional Hub Position

Connects regional data flows, cloud platforms and high-vale digital services

Industry Snapshot



Regional Context Johor's growth creates both competition and complementarity in regional data center development.

Policy evolution: From capacity expansion to selective and sustainable growth

2019 — Developments Pause

New data centre developments on State land were paused due to resource constraints.

2022 — Selective Restart

DC-CFA resumed capacity allocation based on efficiency, decarbonization, connectivity and economic value.

2025 — Stricter Sustainability Requirements

DC-CFA2: PUE ≤1.25 and ≥50% eligible green energy for new capacity.

2026 — Expanding Efficiency Regulation

Digital Infrastructure Bill establishes sustainability requirements for existing and new data centers; implementation expected **from 2027**.

Growing Data Centre Demand in a Carbon-Constrained Power System

Electricity & Climate Context

6.5 TWh (2024)
Data Center electricity consumption
>10% of Singapore electricity use

Material and growing electricity demand

94% (2024)
Natural gas in electricity generation
Grid emission factor: 0.402 kgCO2/kWh

Electricity remains carbon-intensive

2050
National net-zero target
45-50 MtCO2e by 2035

Future growth must prioritize sustainability and value over volume

The Decarbonization Challenge

Growing 24/7 electricity demand must be met with reliable, lower-carbon supply under the following constraints

01 Limited Low-Carbon Electricity Supply

- Solar limited to ~10% of projected 2050 electricity demand
- Cross-border import constraints
- Uncertain deployment of emerging technologies

02 24/7 Power Reliability & Matching

- Continuous electricity demand
- Need for firm generation and grid backup
- Hourly low-carbon supply matching challenges

03 Land & Water Constraints

- Limited land for expansion and generation
- Median water usage effectiveness: ~2.1 m^3/MWh
- Need for integrated site development

04 Tropical Cooling Challenges

- Hot and humid climate
- Cooling accounts for up to 40% of energy use
- Increasing AI rack power density

Research Question

To what extent can a grid-connected data centre in Singapore reduce its annual electricity-related emissions, and at what cost?

Data Center Emissions: Sources, Challenges & Research Boundary

Upstream Activities (Value Chain)

Scope 3
Other indirect emissions

Sources
- Materials & equipment
- Cooling / UPS production
- Fuel & electricity production

Key Challenges
- High embodied carbon
- Limited supplier visibility
- Dependence on upstream decarbonization

Data Center Operation (Operational Phase)

Scope 1
Direct emissions

Sources
- Backup generators
- Refrigerant leakage
- Other on-site fuel use

Key Challenges
- Low-carbon backup options
- Reliability and cost
- Refrigerant

Scope 2
Indirect emissions from purchased electricity

Sources
- Electricity for IT equipment
- Cooling systems
- Supporting infrastructure

Key Challenges
- Continuous and growing electricity demand
- High cooling load in a tropical climate
- Limited low-carbon electricity

Primary focus of this study

Downstream Activities (Value Chain)

Scope 3
Other indirect emissions

Sources
- Colocation/leased assets
- Customer use of services
- End-of-life treatment

Key Challenges
- Complex boundaries
- Limited data
- Limited influence on end use

Research Focus: Electricity-related emissions

Supply-side analysis
Reduce the carbon intensity of electricity supply

+

Demand-side analysis
Reduce facility electricity demand

→

Integrated Assessment
Combine reduction potential cost, reliability and implementation feasibility

Note: the study focus on electricity-related emissions, including selected life-cycle factors for supply-side comparison. A full scope 1-3 inventory is outside the study boundary

Research Design Overview

Research Objective | To develop and evaluate an integrated decarbonisation strategy for a grid-connected data centre in Singapore, considering emissions reduction potential, costs and operational constraints.

Research Scope

Reference Facility & Location

- Hypothetical 100 MW IT (Jurong Island)
- 85% load factor
- Annual PUE 1.25

Assessment Focus

- Energy efficiency & electricity demand
- Electricity-related emissions
- Electricity costs & abatement costs
- Operational reliability & feasibility

Key Boundaries

- Electricity-related emissions (incl. selected supply life-cycle emissions)
- Annual allocation; illustrative hourly matching
- No detailed dispatch, transmission or market modelling

Methodology and Research Framework | Scenario-based Techno-Economic Assessment

1 Establish Baseline



Bottom-up Electricity Demand Estimation

$$E_{DC} = P_{IT} \times LF \times 8760 \times PUE$$

930.8 Gwh/year | 374 Kt CO2/year | S$186.2 M/year

2 Assess Decarbonization Options

Supply Side: Low-carbon Electricity Supply

Technology Screening → Three pathways

- A Rooftop Solar PV
- B 77 MWe SMR
- C 60 MWe Ammonia

Demand Side: Energy Efficiency Optimization

Three Energy Efficiency Measures

- A Predictive Cooling Control
- B Set[pomt Optimization
- C Hybrid Air–Liquid Cooling

3 Identify Strategic Gaps

- Efficiency & Growth — Gaps 1-2
- Supply & Security — Gaps 3-4
- Accounting & Impact — Gaps 5-6

4 Integrated Assessment

Integrated Emissions & Cost Assessment → Safety, Reliability & Implementation

Six Configurations | Sensitivity Analysis | Illustrative Hourly CFE Matching

Final Output | Integrated Decarbonization Strategy & Implementation Recommendations

Types of Renewable and Low-Carbon Energy Sources

Category	Option
Domestic renewable or low-carbon generation	Solar PV
	Wind
	Hydropower
	Geothermal
	Nuclear (SMR)
Imported fuels and electricity	Low-carbon ammonia
	Hydrogen
	Gas with carbon capture (CCS)
	Regional low-carbon electricity

Domestic renewable generation

(a) Solar PV
HDB rooftop arrays, Punggol



(b) Wind
Block Island offshore farm, USA



(c) Hydropower
Hydroelectric plant, Nevada, USA



(d) Geothermal
Wairakei plant, New Zealand



(e) SMR
Artist's impression, NuScale plant



Imported fuels and electricity



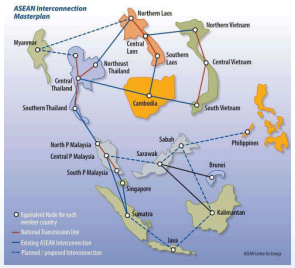
Ammonia



Hydrogen



Gas with carbon capture (CCS)



Imported Electricity

Screening of Energy Options in Singapore

	Resource availability	Technological maturity	Scalability	Reliability	Economic feasibility	Infrastructure requirements	Deployment timeline	**Recommended implementation**
Solar PV	About 1,580 kWh/m²/yr; land limited	Commercial	About 10% of 2050 needs	Variable; daytime only	S$85/MWh, below the grid price	Rooftop; no battery needed	Now; 3 GWp target by 2030	Implement: short term (Scenario A)
Wind	2 to 3 m/s	Commercial elsewhere	None in Singapore	Low output expected	No viable resource	Offshore sites limited by shipping	None	Exclude from domestic deployment
Hydropower	No rivers or head	Commercial elsewhere	None in Singapore	Firm where available	No suitable sites	No river system	None	Exclude from domestic deployment
Geothermal	70 to 122 °C at 1 to 2 km	Enhanced systems pre-commercial	Unproven	Firm if developed	US$140/MWh for US enhanced systems	Wells to about 5 km	No timeline announced	Continue assessment: long term (not modelled)
Nuclear (SMR)	Compact site; fuel imported	Early projects costly	Modular; late 2030s at earliest	93% capacity factor	S$230/MWh (114 to 281)	National nuclear framework needed	Late 2030s at the earliest	Continue assessment: long term (Scenario B)
Low-carbon ammonia	Imported fuel; established supply chain	Gas-turbine firing at demonstration scale	55 to 65 MW project; fuel-limited	Dispatchable	S$342/MWh (260 to 531)	Imported fuel, storage and bunkering	From about 2030 (design stage)	Develop for implementation: medium term (Scenario C)
Hydrogen	Imported fuel	New gas units built hydrogen-ready	Up to half of power by 2050 (strategy)	Dispatchable gas turbines	Not assessed	Hydrogen-ready plant and fuel supply	Future fuel	Continue assessment: long term (not modelled)
Gas with carbon capture (CCS)	No suitable storage sites	Tested in roadmap pilots	Limited by storage geology	Firm gas plant	About 21% of plant output	Overseas storage needed	No domestic timeline	Exclude from domestic deployment (not modelled)

Favourable Partial Unfavourable Uncertain

Scenario A: Solar PV

Singapore's average solar irradiation is ~ 1,580 kWh/m²/year. However, land constraints limit large-scale solar deployment.

Scenario Overview



This scenario models rooftop solar PV installation for the 100 MW reference data centre, using Digital Loyang II as a reference for building dimensions.

Rooftop area and installed capacity

Estimated gross floor area	68,000 m²
Available rooftop area	13,600 m²
Rooftop utilization factor	60%
Installed capacity density	0.14 kWp/m²

Installed PV capacity = 13,600 × 0.60 × 0.14 = **1,142 kWp ≈ 1.14 MWp**

Annual Electricity Generation

1,142.4 × 1.385 =
1,582 MWh/year

Contribution & Peak Output

0.17%
of the annual electricity demand

0.57MW
Peak output (vs. average facility demand of 106.25 MW)

Indicative Electricity Cost

S$85/MWh
Rooftop solar PV

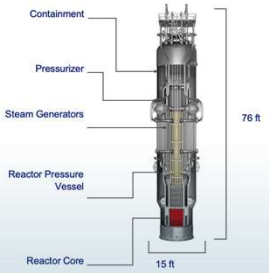
S$200/MWh
Grid electricity

Key takeaway: At S$85/MWh, rooftop solar PV is cheaper than grid electricity (S$200/MWh), but its contribution is constrained by available rooftop area.

Scenario B: Nuclear SMR

Small Modular Reactors (SMRs) are factory-built nuclear fission reactors (up to ~300 Mwe) with passive safety systems.

Scenario Overview



- Pressurized-water SMR: fission heat -> steam -> turbine-generator
- Provides potential continuous low-carbon electricity

Key Parameters

Module capacity (NuScale six-module design)	**77 MWe**
Capacity Factor	**93%**

Annual Electricity Generation

$77 \times 0.93 \times 8{,}760 =$
627,304 MWh/year

Contribution

67.4%
of the annual electricity demand

Singapore's Nuclear Readiness & Deployment

- **> S$150 million** invested in nuclear research
- IAEA INIR Phase 1 mission planned for 2027
- IAEA Milestones: at least **10-15 years** to first plant operation
- SMR is a **longer-term option**, potentially from the **late 2030s**

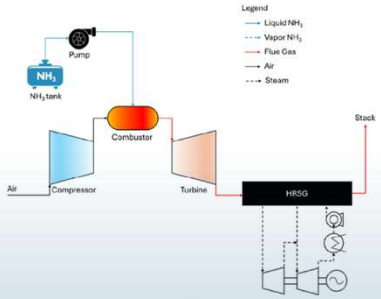
Key takeaway: SMRs provide firm low-carbon electricity, but deployment remains dependent on national approval, safety regulations, infrastructure readiness and economic feasibility.

Scenario C: Low-Carbon Ammonia-Fired Generation



Ammonia emits no carbon during combustion (under ideal conditions) $4NH_3 + 3O_2 \rightarrow 2N_2 + 6H_2O$

Scenario Overview



- A 60 MWe GTCC plant in the new-energy area of Jurong Island
- Uses direct liquid-ammonia combustion

Key Parameters

Net Efficiency	**47.7%**
Capacity Factor	**85%**

Annual Electricity Generation

60 x 0.85 x 8,760 =
446,760 MWh/ yr

Fuel Demand

- Specific fuel consumption (SFC) = **0.4014 tNH3/ MWh**
- **Annual fuel demand** = **179,349 t/ yr = 491 t/ day**

Contribution

48%
of the annual electricity demand

Key takeaway: Ammonia has potential as a low-carbon fuel for dispatchable energy, but concerns over cost and carbon footprint is dependent on its supply chain

Scenario C: Low-Carbon Ammonia-Fired Generation

Challenges Faced by NH_3 Fired Generation

Poor Combustion Kinetics

Combustion Stability Risk

- Flame instability, blowout or misfiring
- Potential turbine trips

Incomplete Combustion

- NH_3 waste and reduced thermal efficiency
- Toxic NH_3 slip in exhaust gas

Emissions Challenge

- NO_x emissions (NO, NO_2, N_2O)
- N_2O formation (potent greenhouse gas)



Low Heating Value

Higher Fuel Requirement

- More NH_3 needed for the same output

Equipment Design Complexity

- Increased fuel throughput and handling requirements

**Compared with natural gas and H_2*

Scenario C: Low-Carbon Ammonia-Fired Generation



Supply Chain & Life-cycle Emission

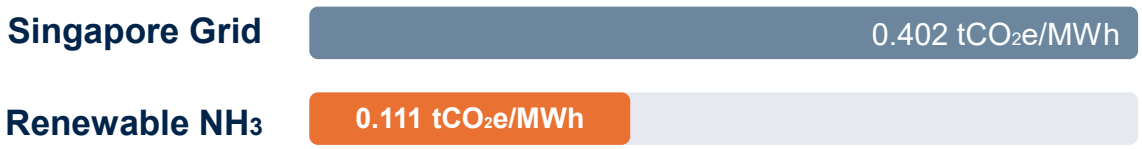
Ammonia's climate benefits depend heavily on its upstream supply chain

Emission Factor Calculation

$$EF_{NH3} = SFC_{NH3} \times I_{NH3} + EF_{N2O}$$

Emission Factor Comparison

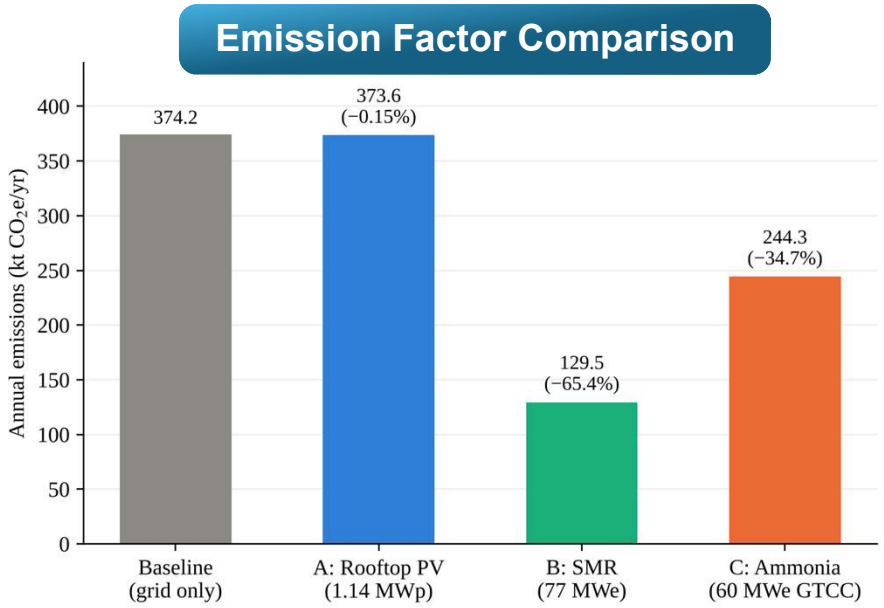
Based on the central, middle-ground scenario (accounting for production, embodied electricity, shipping)



~72%
reduction in emissions

Comparative Emission Assessment of All Three Scenarios

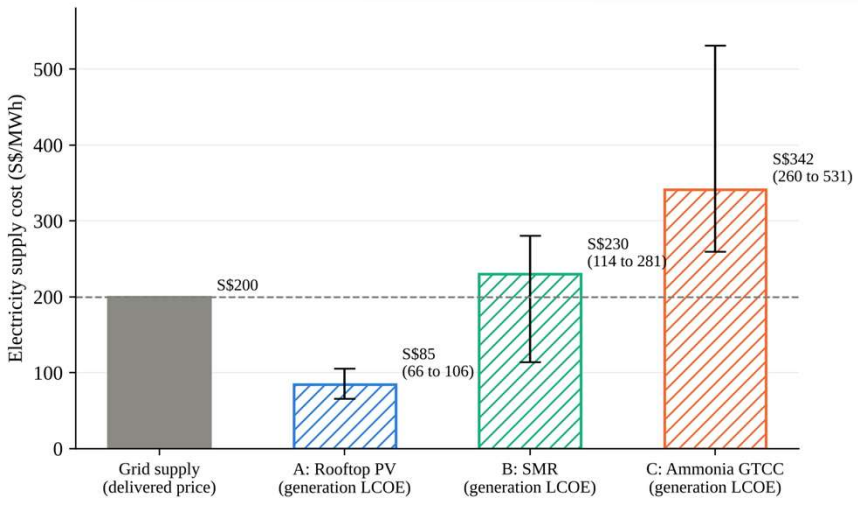
Scenario	Generation	Share of demand	Residual grid usage
A: Rooftop PV, 1.14 MWp	1,582	0.17%	929,168
B: SMR, 77 MWe	627,304	67.4%	303,446
C: Ammonia GTCC, 60 MWe	446,760	48.0%	483,990
	PV 1,582; SMR 627,304; ammonia 301,864	100%	0



Techno-Economic Assessment

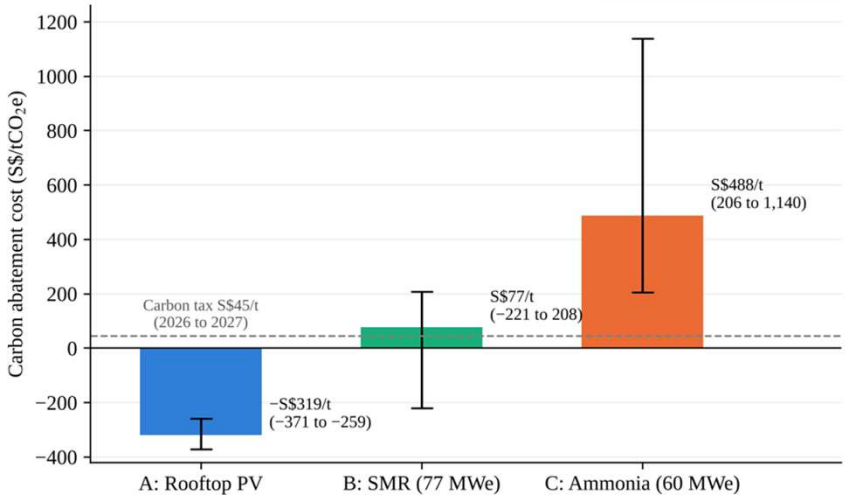
Electricity Supply Demand

- Grid LCOE: $200/ MWh
- LCOE of PV is lowest at $85/ MWh
- LCOE of SMR is the middle ground at $230/ MWh
- LCOE of NH_3 is the highest at $342/ MWh due to high fuel cost



Carbon Abatement Cost

- PV saves S$0.18 million a year at a net saving of S$319/$tCO_2e$.
- SMR adds S$18.8 million at S$77/tCO_2e, from a net saving at the subsidised cost to S$208/$tCO_2e$
- NH_3 plant adds S$63.4 million at S$488/tCO_2e



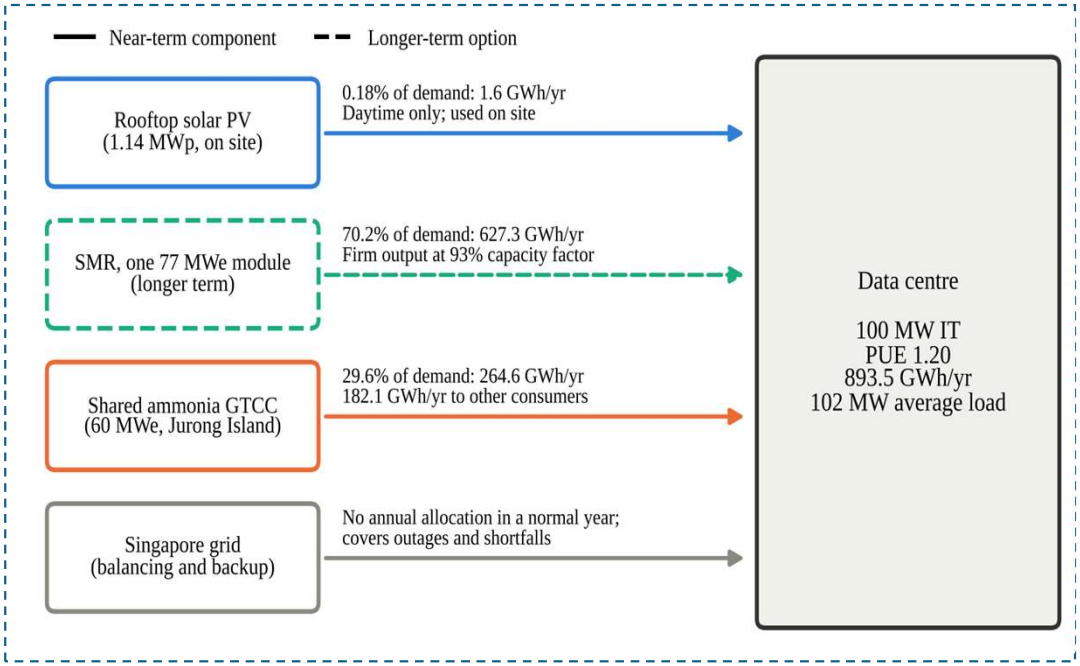
Proposed Approach

The 3 technologies are complementary

- Rooftop PV is mature and cheaper than the grid but capped at 0.17% of demand.
- Renewable NH3 projected to be available this decade and cuts life-cycle emissions by almost 35%
- SMRs deliver the largest and cheapest supply-side abatement with firm output, but has the lowest clarity on availability

Recommended Approach

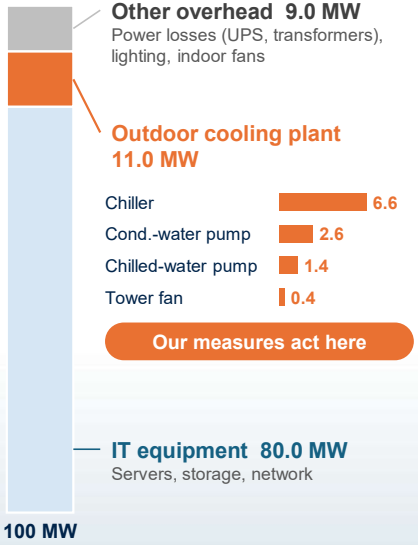
- Install PV now to maximise available gains
- Buy renewable NH3 when available
- Add SMR capacity once available, progressively reducing NH3 usage as needed



Hardware + Software Save ~16 MW of Every 100 MW Supplied

Where 100 MW of supplied power goes, and how it is saved

01 Baseline 100 MW in



PUE 1.25 baseline
= 1 + 0.14 cooling + 0.11 other

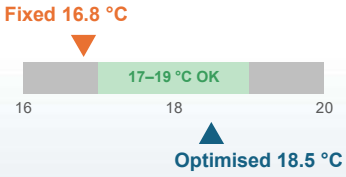
02 Problem Waste

Hardware

Air carries all the heat
Server and room fans run hard; chilled water must be kept cold

Software

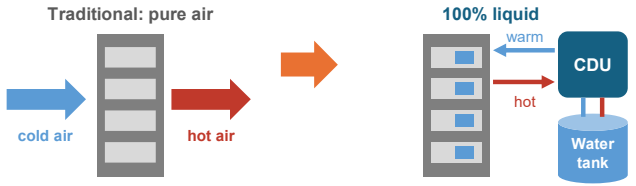
Fixed setpoints over-cool



91.8%
of the time colder than needed

03 Solution Hardware + Softwawre

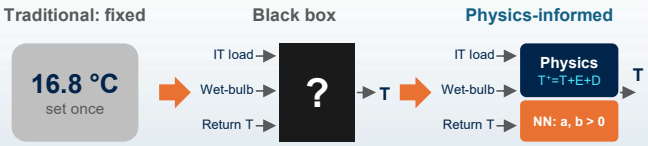
Hardware Pure air → 100% liquid cooling



Fans move 100% of the heat

Cold plates, CDU + tank; no fans

Software Fixed → black box → physics-informed

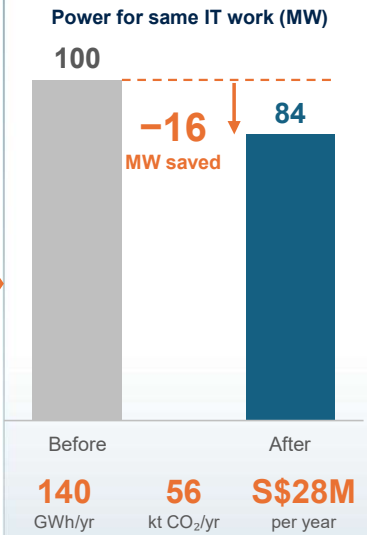


No feedback; over-cools

Data only; fails on new load

Physics built in; safe

04 Result -16MW



140	56	S$28M
GWh/yr	kt CO_2/yr	per year

Final PUE 1.25 → 1.05

Summary Combining hardware and software on the demand side: 100 MW → 84 MW for the same IT work, PUE 1.25 → 1.05.
Industry proof Google Singapore (PUE 1.12) does the same: hardware = high-temperature operation, efficient power supplies, custom servers; software = AI cooling control.

Caveats: 1.05 is a design target, not a measured Singapore result; software results come from a 30-day calibrated co-simulation at NUS STDCT.

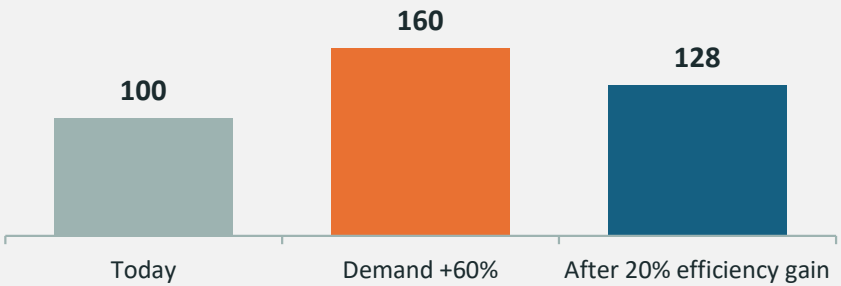
Sources: Chen, Chui & Lee (A-PCNN; PIML-MPC, NUS STDCT); ORNL Frontier (liquid-cooled, PUE ≈ 1.05); Google data-centre efficiency data (Singapore TTM PUE 1.12, Q2 2026). Scaling: 100 MW continuous, 0.402 kgCO_2/kWh, S$0.20/kWh.

The Efficiency Challenge in AI-Driven Data Centres

AI workloads (LLMs, generative AI, HPC) are raising compute demand much faster than traditional applications.

Gap 1: the efficiency paradox (rebound effect)

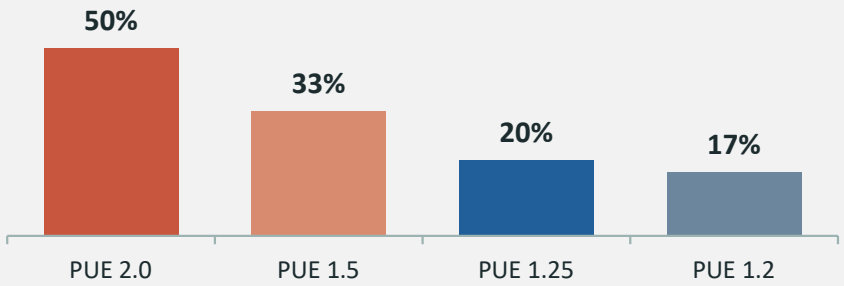
Electricity use, index (today = 100)



A 20% efficiency gain against 60% AI demand growth still raises electricity use by 28% (100 × 1.6 × 0.8).

Gap 2: diminishing returns from PUE

Maximum facility energy that overhead removal can save (1 − 1/PUE)



Facility overhead cannot fall below PUE 1.0. Decarbonising therefore needs efficient AI hardware and workloads as well as efficient facilities.

Takeaway Efficiency improvements are necessary, but AI-driven growth may offset much of the energy savings.

Renewable and Imported Energy Limitations

Gap 3: Renewable energy mismatch

- Data centres need continuous 24/7 power; solar is intermittent and weather dependent.
- Firm supply also needs large-scale storage, smart-grid management and firm backup generation, at very high availability.

45% of demand matched hour by hour, even if solar generation equals 100% of annual demand

Renewable capacity ≠ firm power capacity

Gap 4: Dependence on imported energy

- Singapore relies increasingly on imported low-carbon electricity, hydrogen and ammonia.
- Risks: geopolitical uncertainty, cross-border transmission failures, export-country policy changes and price volatility.

S$51/MWh added to ammonia power cost by each US$100/t move in fuel price

Example: Indonesia cut pipeline gas exports by about 14% for 18 days in Oct 2026

Takeaway Decarbonisation success increasingly depends on external energy systems and regional cooperation.

Defining Net Zero and Measuring Real Progress

Gap 5: Ambiguous definition of net zero

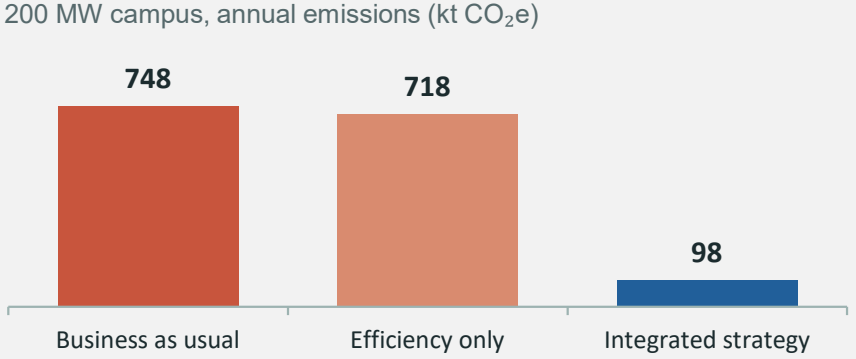
- A true assessment covers Scope 1, Scope 2 and Scope 3 emissions.
- Annual renewable matching is not 24/7 carbon-free energy: a facility can claim net zero while using fossil electricity in some hours.

5% hourly carbon-free energy in Google's Singapore cloud region in 2025

Annual matching versus 24/7 carbon-free energy

Gap 6: Efficiency over absolute consumption

- Policy improves PUE, IT efficiency and carbon intensity, but net zero is set by total electricity and total emissions.



Takeaway Net zero requires stronger carbon accounting and policies that target absolute emissions reduction.

Integrated Emission Reduction & Implementation Consideration

Integrated emissions reduction pathway

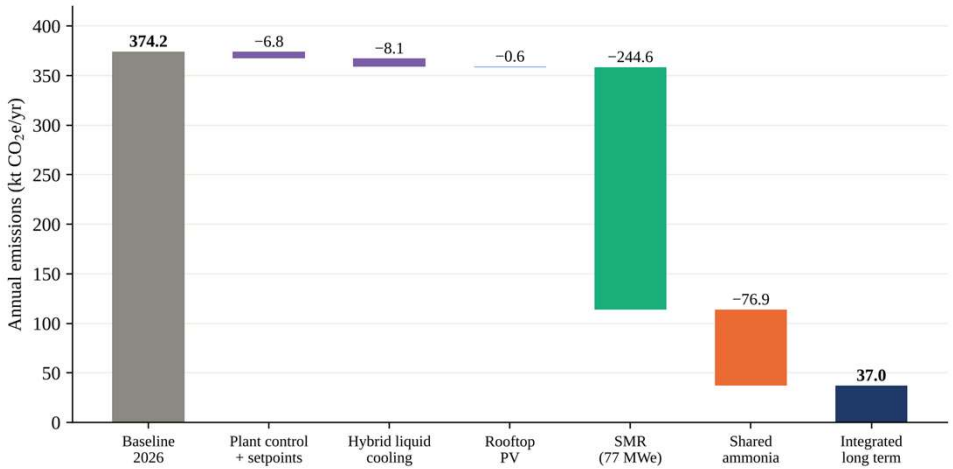


Figure 5.4. Annual emissions of the reference data centre from the 2026 baseline to the integrated long-term configuration, by measure

90.1% emission reduction (from 374.2 kt to 37.0 kt CO_2e/yr)

Safety, Reliability & Implementation

Measure	Hazard	Engineering control
PIML-MPC plant control	Setpoints outside the thermal band; model drift; cyber intrusion	Physics constraints, hard limits in the BMS and automatic fallback to fixed setpoints; segregated control network with read-only optimiser paths
Warmer halls (SS 697)	Server faults and higher IT fan power	Operate at the measured total-power optimum; agree the ASHRAE envelope in tenant contracts [12]
Hybrid liquid cooling (SS 726)	Coolant leaks, corrosion, biofilm, floor loading	Leak detection and isolation, fluid-quality monitoring, compatible piping and structural checks [16]
Cooling towers	Legionella in warm recirculating water	Continuous biocide treatment, routine sampling and drift elimination
Chillers	Fugitive leakage of high-GWP refrigerant (Scope 1)	Leak detection, annual audits and low-GWP refrigerant at chiller renewal
UPS batteries	Thermal runaway and fire	Fire-rated separation, cell-level monitoring and suppression
Ammonia GTCC and fuel handling (Jurong Island)	Ammonia toxicity and corrosivity; leaks; flame instability; NO_x, N_2O and ammonia slip	Gas detection with automatic isolation, ventilation, secondary containment and separation distances; staged low-NO_x combustion with SCR and continuous emissions monitoring [58], [34]; emergency plans coordinated with neighbouring plants
SMR (long term)	Nuclear safety, security and waste	National regulatory framework under the IAEA Milestones Approach, prepared through NEA safety studies and the 2027 INIR mission [61], [19]; the operator contracts for output only

Implementation requires safety, operational reliability, and regulatory readiness

Implementation Roadmap



Cooling Optimisation and Rooftop Solar

Sub-metering, PIML-MPC roll-out, raise operating temperatures, install rooftop and facade PV

PUE 1.227
with energy and cost savings

Phase 1

Hybrid Liquid Cooling

Deploy hybrid air and direct-to-chip liquid cooling in all AI halls (SS 726)

PUE 1.20
With further energy and cost savings

Phase 2

Ammonia Electricity Integration

Contract certified ammonia electricity from the Jurong Island plant

Up to 446.8 GWh/yr
of low-carbon supply

Phase 3

Potential SMR Integration

Contract SMR output after a national decision and reduce ammonia to residual demand

627.3 GWh/yr
of firm, low-carbon supply

Phase 4

Phase 5

Integrated operation and verification (throughout)
Operate the portfolio with grid balancing and report metered emissions and hourly CFE

Key Stakeholders for Data Center Decarbonization

Industry

PSA
Keppel
STTelemedia Global Data Centres
EQUINIX
nxera



Policy and Regulatory

Energy Market Authority – Our Clean Energy Future
jtc
METI Ministry of Energy, Trade and Industry Singapore
MPA Singapore
EDB Singapore

Research Institutes

NUS National University of Singapore
Agency for Science, Technology and Research Singapore
SNRSI Singapore Nuclear Research and Safety Institute 16 Prince George's Park, Singapore 118415
IAEA
National Environment Agency Safeguard · Nurture · Cherish

Reflection and Key Learning Points

What we learned

Net zero for data centres is a multidimensional challenge:

- **Energy efficiency**
- **Renewable integration**
- **Low-carbon power imports**
- **Policy and regulatory frameworks**

Key insights

- Efficiency improvements alone cannot achieve net zero if AI demand keeps growing rapidly.
- Reliability, affordability and sustainability must be balanced at the same time.
- Singapore's land constraints make regional energy partnerships essential.

Team reflection

We developed a stronger understanding of:

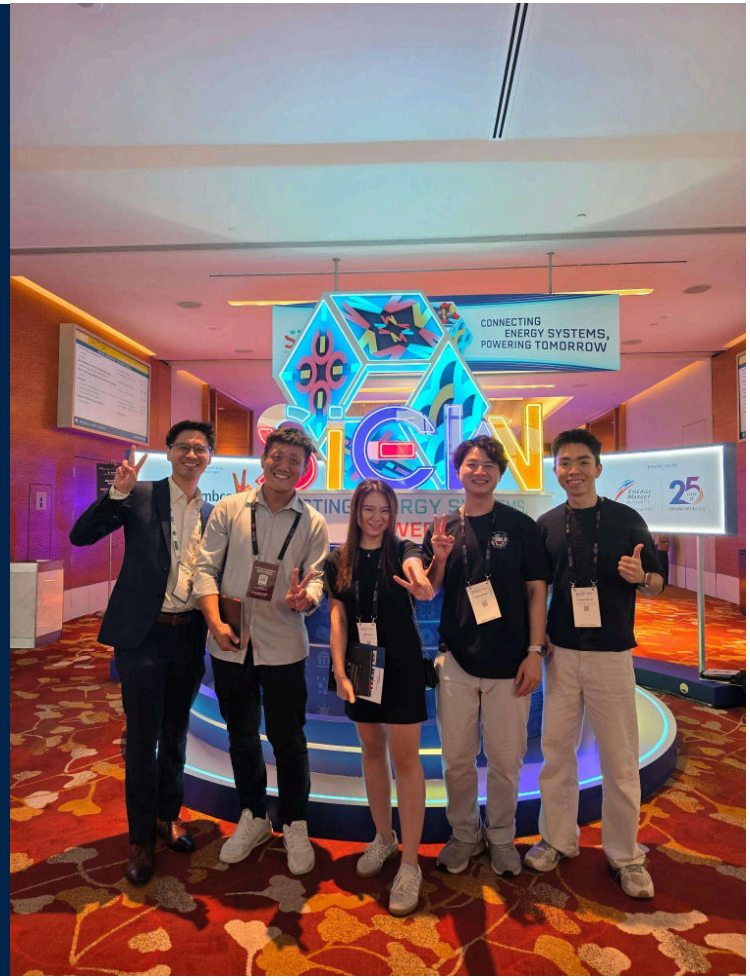
- Data-centre energy systems
- AI energy impacts
- Energy transition strategies
- Sustainability trade-offs

We saw why technical, economic and policy perspectives must be integrated when evaluating decarbonisation pathways.

Final reflection *Singapore's strategy is among the world's most comprehensive. Future success depends on managing growing AI demand while accelerating access to reliable low-carbon energy.*

Thank You!

Questions & Discussion



Types of Renewable and Low-Carbon Energy Sources

Category	Option
Domestic renewable or low-carbon generation	Solar PV
	Wind
	Hydropower
	Geothermal
	Nuclear (SMR)
Imported fuels and electricity	Low-carbon ammonia
	Hydrogen
	Gas with carbon capture (CCS)
	Regional low-carbon electricity

(a) Solar PV
HDB rooftop arrays, Punggol



(b) Wind
Block Island offshore farm, USA



(c) Hydropower
Hydroelectric plant, Nevada, USA



(d) Geothermal
Wairakei plant, New Zealand



(e) SMR
Artist's impression, NuScale plant





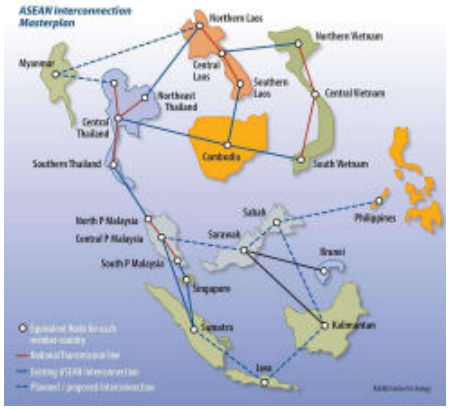
Ammonia



Hydrogen



Gas with carbon capture (CCS)



Imported Electricity

Screening of energy options in Singapore

	Resource availability	Technological maturity	Scalability	Reliability	Economic feasibility	Infrastructure requirements	Deployment timeline	**Recommended implementation**
Solar PV	About 1,580 kWh/m²/yr; land limited	Commercial	About 10% of 2050 needs	Variable; daytime only	S$85/MWh, below the grid price	Rooftop; no battery needed	Now; 3 GWp target by 2030	Implement: short term (Scenario A)
Wind	2 to 3 m/s	Commercial elsewhere	None in Singapore	Low output expected	No viable resource	Offshore sites limited by shipping	None	Exclude from domestic deployment
Hydropower	No rivers or head	Commercial elsewhere	None in Singapore	Firm where available	No suitable sites	No river system	None	Exclude from domestic deployment
Geothermal	70 to 122 °C at 1 to 2 km	Enhanced systems pre-commercial	Unproven	Firm if developed	US$140/MWh for US enhanced systems	Wells to about 5 km	No timeline announced	Continue assessment: long term (not modelled)
Nuclear (SMR)	Compact site; fuel imported	Early projects costly	Modular; late 2030s at earliest	93% capacity factor	S$230/MWh (114 to 281)	National nuclear framework needed	Late 2030s at the earliest	Continue assessment: long term (Scenario B)
Low-carbon ammonia	Imported fuel; established supply chain	Gas-turbine firing at demonstration scale	55 to 65 MW project; fuel-limited	Dispatchable	S$342/MWh (260 to 531)	Imported fuel, storage and bunkering	From about 2030 (design stage)	Develop for implementation: medium term (Scenario C)
Hydrogen	Imported fuel	New gas units built hydrogen-ready	Up to half of power by 2050 (strategy)	Dispatchable gas turbines	Not assessed	Hydrogen-ready plant and fuel supply	Future fuel	Continue assessment: long term (not modelled)
Gas with carbon capture (CCS)	No suitable storage sites	Tested in roadmap pilots	Limited by storage geology	Firm gas plant	About 21% of plant output	Overseas storage needed	No domestic timeline	Exclude from domestic deployment (not modelled)

Favourable Partial Unfavourable Uncertain

Scenario A: Solar PV

Singapore's average solar irradiation is approximately 1,580 kWh/m²/year. However, limited land availability restricts large-scale solar deployment. This scenario models rooftop solar PV installation for the 100 MW reference data centre, using Digital Loyang II as a reference for building dimensions.



Rooftop area and installed capacity:

- Estimated gross floor area = 68,000 m²
- Available rooftop area = 13,600 m²
- Rooftop utilisation factor = 60%
- Installed capacity density = 0.14 kWp/m²

Installed PV capacity = 13,600 × 0.60 × 0.14 = **1,142 kWp ≈ 1.14 MWp**

Annual electricity generation:

Energy produced = 1,142.4 × 1.385 = **1,582 MWh/year**

The solar PV system supplies only **0.17% of the data centre's annual electricity demand**, with a peak output of approximately 0.57 MW against an average facility demand of 106.25 MW.

At S$85/MWh, rooftop solar PV is cheaper than grid electricity (S$200/MWh), but its contribution is constrained by available rooftop area.

Scenario B: Nuclear SMR

Small Modular Reactors (SMRs) are factory-built nuclear fission reactors with electrical capacities of up to approximately 300 MWe and passive safety systems. In a pressurised-water SMR, nuclear fission generates heat, which is transferred through a steam generator to produce steam that drives a turbine-generator. This scenario adopts a **77 MWe SMR module**, based on the NuScale six-modul reference design, with a **93% capacity factor** to provide continuous low-carbon electricity.

Annual electricity generation:

Energy produced = 77 × 0.93 × 8,760

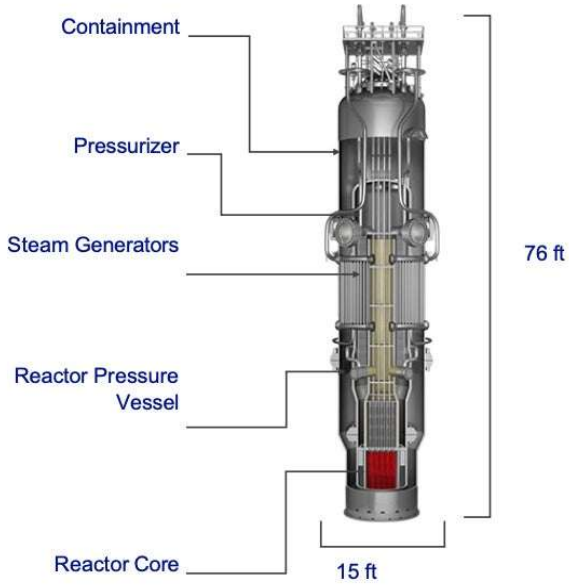
= **627,304 MWh/year**

This generation output is sufficient to meet **67.4% of the modelled data centre' annual electricity demand**.

Singapore's nuclear readiness and deployment:

- Singapore has invested more than S$150 million in nuclear research.
- The IAEA Integrated Nuclear Infrastructure Review (INIR) Phase 1 mission is planned for 2027.
- The IAEA Milestones Approach indicates at least 10–15 years from national consideration to first plant operation.
- SMR electricity supply is therefore considered a longer-term option, potentially from the late 2030s.

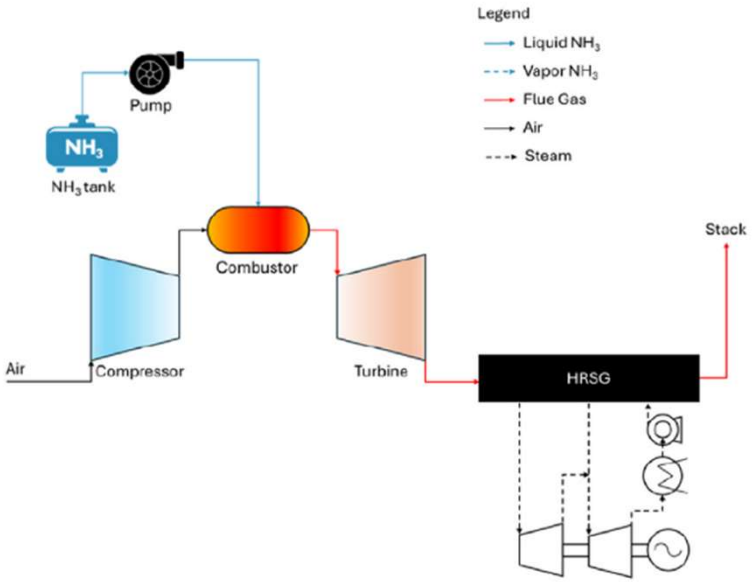
SMRs provide firm low-carbon electricity, but deployment remains dependent on national approval, safety regulations, infrastructure readiness and economic



Scenario C: Low-Carbon Ammonia-Fired Generation

Ammonia emits no carbon during combustion (under ideal conditions):
$4NH_3 + 3O_2 \rightarrow 2N_2 + 6H_2O$

This scenario involves a 60 MWe combined-cycle gas turbine (GTCC) plant in the new-energy area of Jurong Island that also hosts the data centre park. The plant would utilise direct liquid-ammonia combustion.



The plant operates at an estimated 47.7% net efficiency and an 85% capacity factor

Based on plant output, annual fuel demand is calculated

Energy produced = 60 x 0.85 x 8,760 = 446,760 MWh/ yr

Specific fuel consumption (SFC) = 3.6 / (0.477 x 18.8) = 0.4014 tNH3/ MWh

Annual fuel demand = energy production x SFC = 446,760 x 0.4014 = 179,349 t/ yr = 491 t/ day

This production output is able to meet 48% of the modelled data centre's demand

Scenario C: Low-Carbon Ammonia-Fired Generation

Challenges faced by NH3 fired generation

- Poor combustion kinetics compared to natural gas (NG) and H2
  - Leads to flame instability and potential of flame blowout / misfiring that destabilises the generator and can trip the turbine
  - Can lead to incomplete combustion that waste NH3 fuel, degrades thermal efficiency, and releases toxic NH3 into the exhaust steam
  - Can lead to the production of NOx emissions (e.g. NO, NO2, N2O) that are highly potent greenhouse gases
- Low heating value compared to NG and H2 increases the volume of NH3 required, and increases complexity of equipment design to handle increased volume throughput

Scenario C: Low-Carbon Ammonia-Fired Generation

Supply Chain & Life-cycle Emission

- NH3 climate benefits is highly dependent on its upstream supply chain

Using renewable NH3 as a benchmark,

Emission factor (NH3) = [SFC(NH3) x Intensity (NH3)] x EF(N2O)

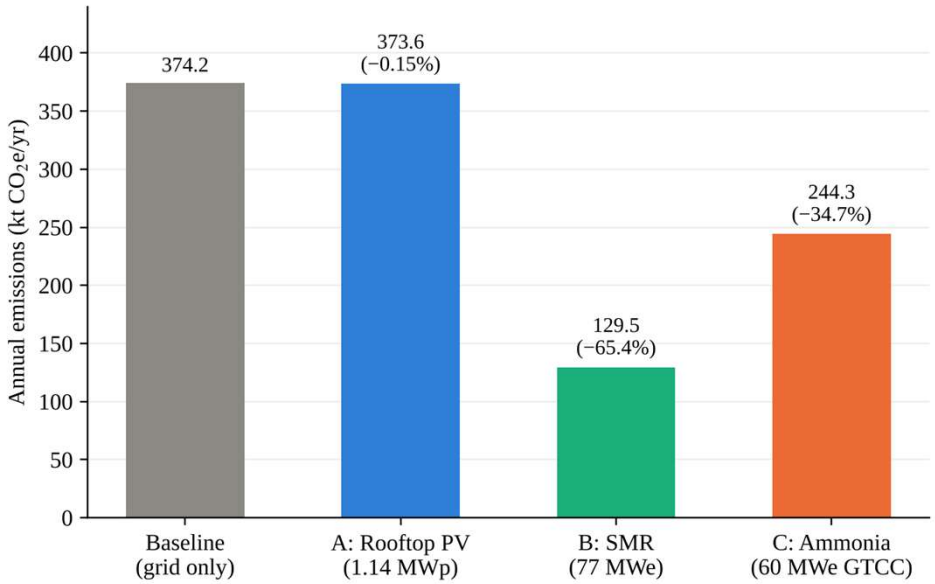
Based on the central, middle-ground scenario (accounting for production, embodied electricity, shipping):

EF(NH3) is 0.111 tCO2e/ MWh Vs Grid factor of 0.402 tCO2e/ MWh

NH3 represents a 72% reduction in emissions

Comparative emission assessment of all 3 Scenarios

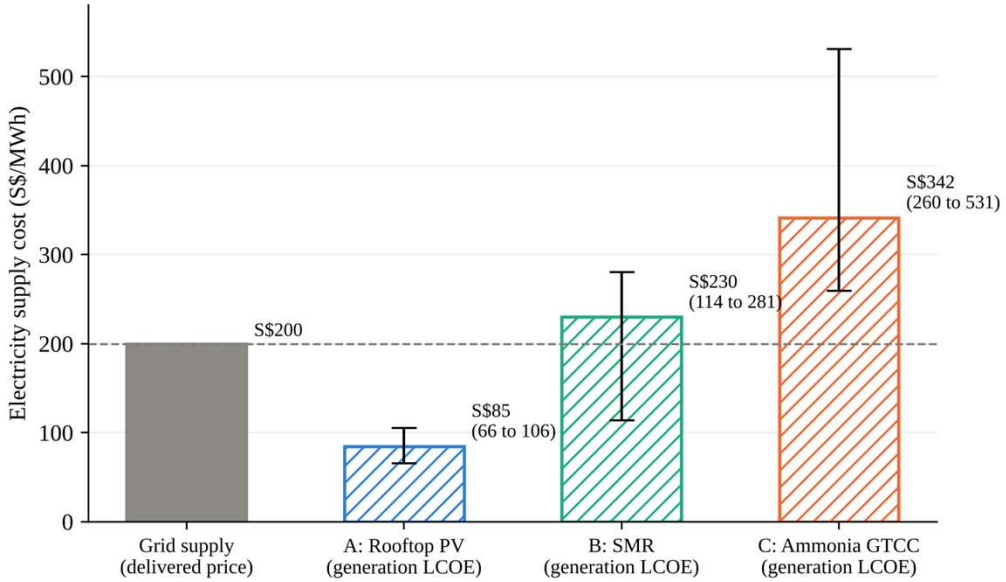
A: Rooftop PV, 1.14 MWp	1,582	0.17%	929,168
B: SMR, 77 MWe	627,304	67.4%	303,446
C: Ammonia GTCC, 60 MWe	446,760	48.0%	483,990
A + B + C portfolio	PV 1,582; SMR 627,304; ammonia 301,864	100%	0



Techno-Economic Assessment

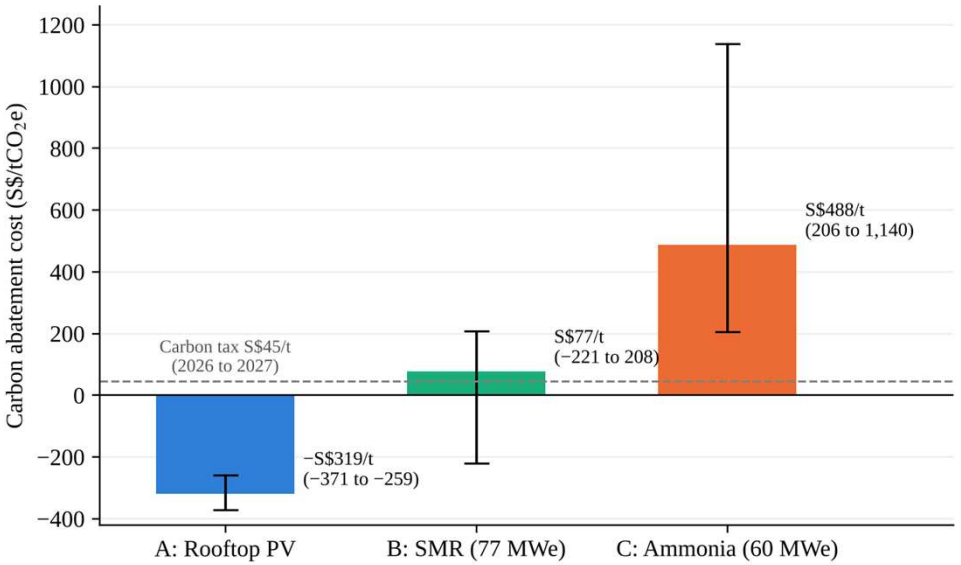
Grid LCOE: $200/ MWh

LCOE of PV is lowest at $85/ MWh
LCOE of SMR is the middle ground at $230/ MWh
LCOE of NH3 is the highest at $342/ MWh due to high fuel cost



Carbon abatement cost

PV saves S$0.18 million a year at a net saving of S$319/$tCO_2e$.
SMR adds S$18.8 million at S$77/tCO_2e, from a net saving at the subsidised cost to S$208/$tCO_2e$
NH3 plant adds S$63.4 million at S$488/tCO_2e



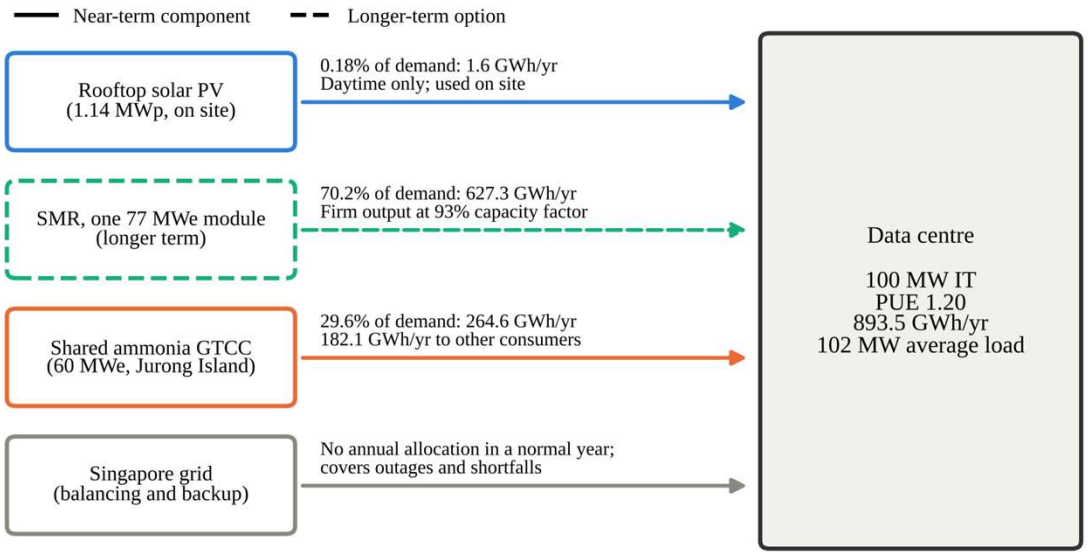
Proposed Approach

The 3 technologies are complementary

- Rooftop PV is mature and cheaper than the grid but capped at 0.17% of demand.
- Renewable NH3 projected to be available this decade and cuts life-cycle emissions by almost 35%
- SMRs deliver the largest and cheapest supply-side abatement with firm output, but has the lowest clarity on availability

Recommended approach

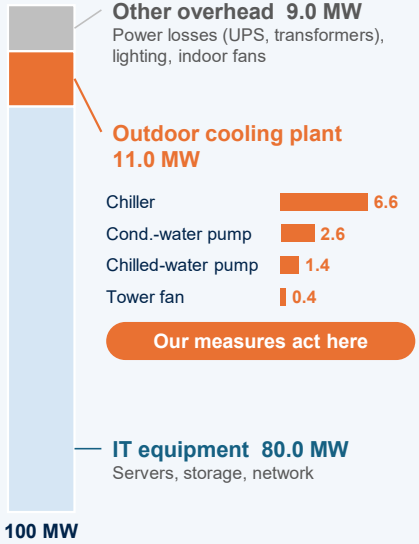
- Install PV now to maximise available gains
- Buy renewable NH3 when available
- Add SMR capacity once available, progressively reducing NH3 usage as needed



Hardware + Software Save ~16 MW of Every 100 MW Supplied

Where 100 MW of supplied power goes, and how it is saved

1 Baseline 100 MW in



PUE 1.25 baseline
= 1 + 0.14 cooling + 0.11 other

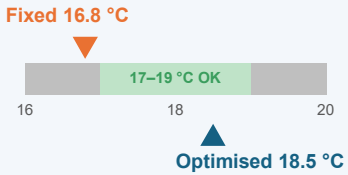
2 Problem Waste

Hardware

Air carries all the heat
Server and room fans run hard; chilled water must be kept cold

Software

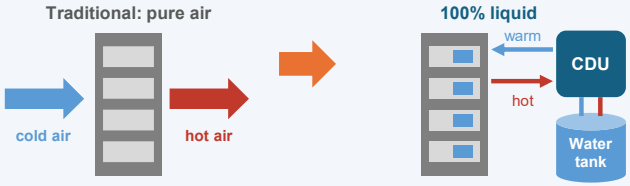
Fixed setpoints over-cool



91.8%
of the time colder than needed

3 Solution Hardware + Software

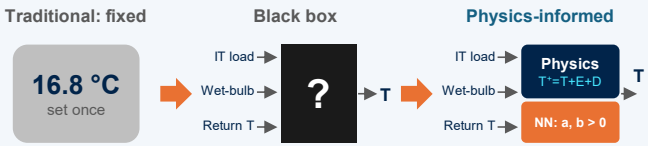
Hardware Pure air → 100% liquid cooling



Fans move 100% of the heat

Cold plates, CDU + tank; no fans

Software Fixed → black box → physics-informed

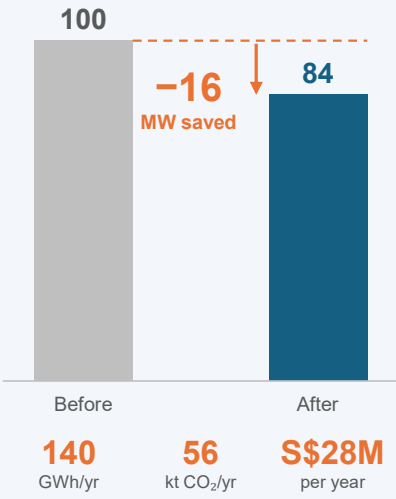


No feedback; over-cools

Data only; fails on new load

Physics built in; safe

4 Result −16 MW



140 GWh/yr	56 kt CO_2/yr	S$28M per year

Final PUE 1.25 → 1.05

Summary Combining hardware and software on the demand side: 100 MW → 84 MW for the same IT work, PUE 1.25 → 1.05.

Industry proof Google Singapore (PUE 1.12) does the same: hardware = high-temperature operation, efficient power supplies, custom servers; software = AI cooling control.

Caveats: 1.05 is a design target, not a measured Singapore result; software results come from a 30-day calibrated co-simulation at NUS STDCT.

Sources: Chen, Chui & Lee (A-PCNN; PIML-MPC, NUS STDCT); ORNL Frontier (liquid-cooled, PUE ≈ 1.05); Google data-centre efficiency data (Singapore TTM PUE 1.12, Q2 2026). Scaling: 100 MW continuous, 0.402 kgCO_2/kWh, S$0.20/kWh.

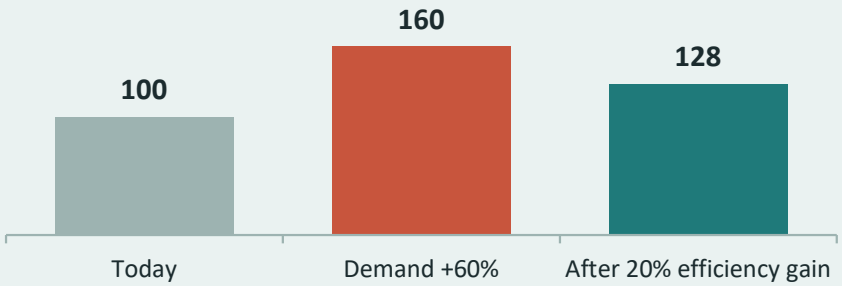
GAP 1 AND GAP 2

The Efficiency Challenge in AI-Driven Data Centres

AI workloads (LLMs, generative AI, HPC) are raising compute demand much faster than traditional applications.

Gap 1: the efficiency paradox (rebound effect)

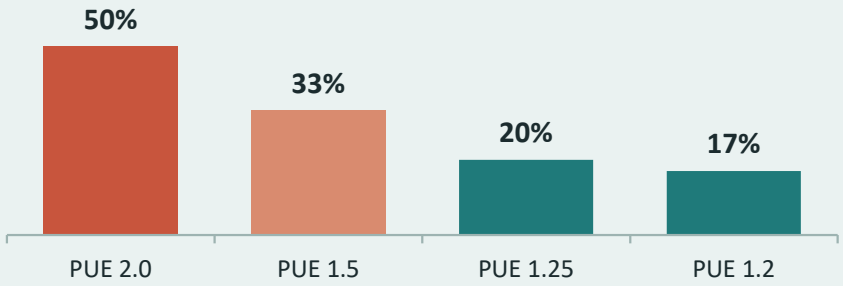
Electricity use, index (today = 100)



A 20% efficiency gain against 60% AI demand growth still raises electricity use by 28% (100 × 1.6 × 0.8).

Gap 2: diminishing returns from PUE

Maximum facility energy that overhead removal can save (1 − 1/PUE)



Facility overhead cannot fall below PUE 1.0. Decarbonising therefore needs efficient AI hardware and workloads as well as efficient facilities.

Takeaway Efficiency improvements are necessary, but AI-driven growth may offset much of the energy savings.

GAP 3 AND GAP 4

Renewable and Imported Energy Limitations

3 Gap 3: Renewable energy mismatch

- Data centres need continuous 24/7 power; solar is intermittent and weather dependent.
- Firm supply also needs large-scale storage, smart-grid management and firm backup generation, at very high availability.

45%

of demand matched hour by hour, even if solar generation equals 100% of annual demand

Renewable capacity ≠ firm power capacity

4 Gap 4: Dependence on imported energy

- Singapore relies increasingly on imported low-carbon electricity, hydrogen and ammonia.
- Risks: geopolitical uncertainty, cross-border transmission failures, export-country policy changes and price volatility.

S$51/MWh

added to ammonia power cost by each US$100/t move in fuel price

Example: Indonesia cut pipeline gas exports by about 14% for 18 days in Oct 2026

Takeaway Decarbonisation success increasingly depends on external energy systems and regional cooperation.

GAP 5 AND GAP 6

Defining Net Zero and Measuring Real Progress

5 Gap 5: Ambiguous definition of net zero

- A true assessment covers Scope 1, Scope 2 and Scope 3 emissions.
- Annual renewable matching is not 24/7 carbon-free energy: a facility can claim net zero while using fossil electricity in some hours.

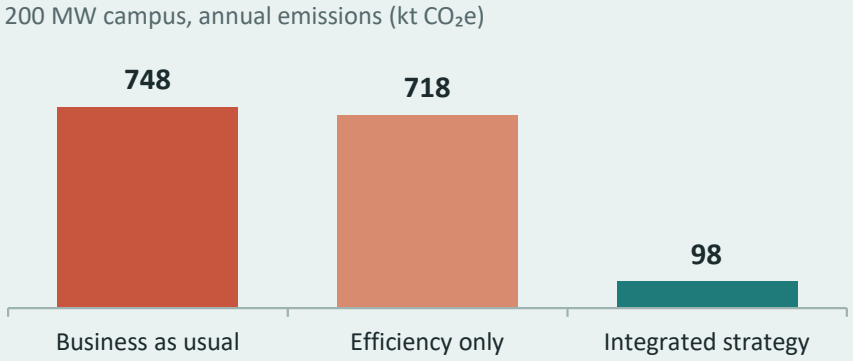
5%

hourly carbon-free energy in Google's Singapore cloud region in 2025

Annual matching versus 24/7 carbon-free energy

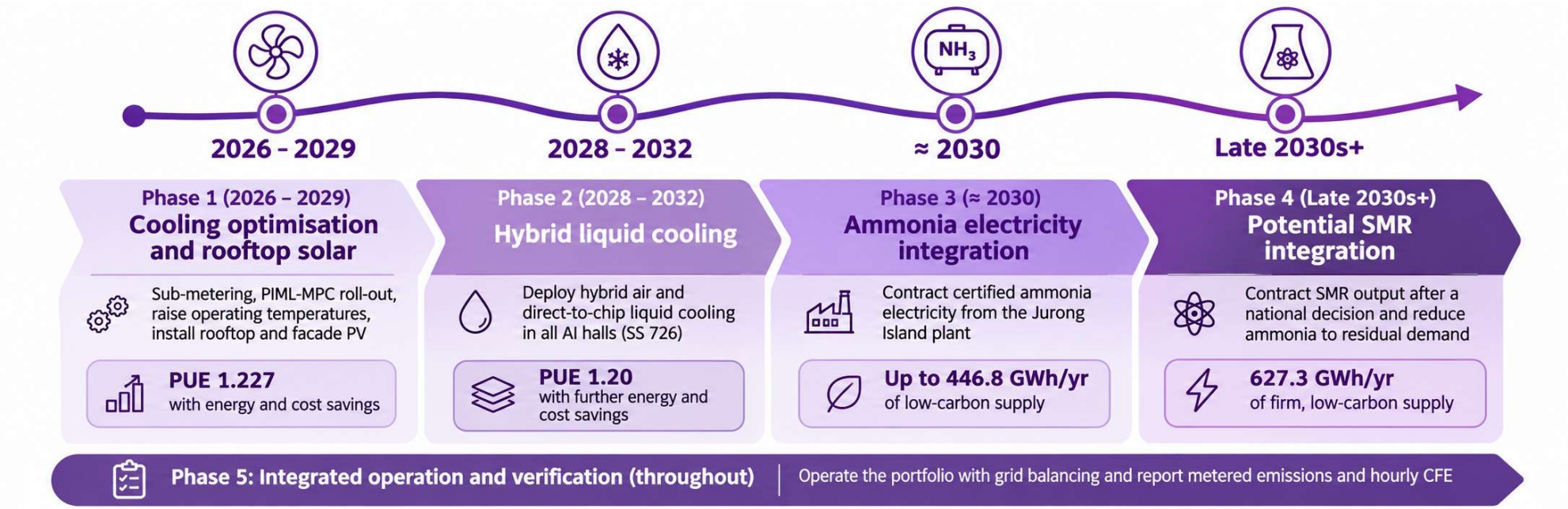
6 Gap 6: Efficiency over absolute consumption

- Policy improves PUE, IT efficiency and carbon intensity, but net zero is set by total electricity and total emissions.



Overall Net zero requires stronger carbon accounting and policies that target absolute emissions reduction.

Implementation Roadmap



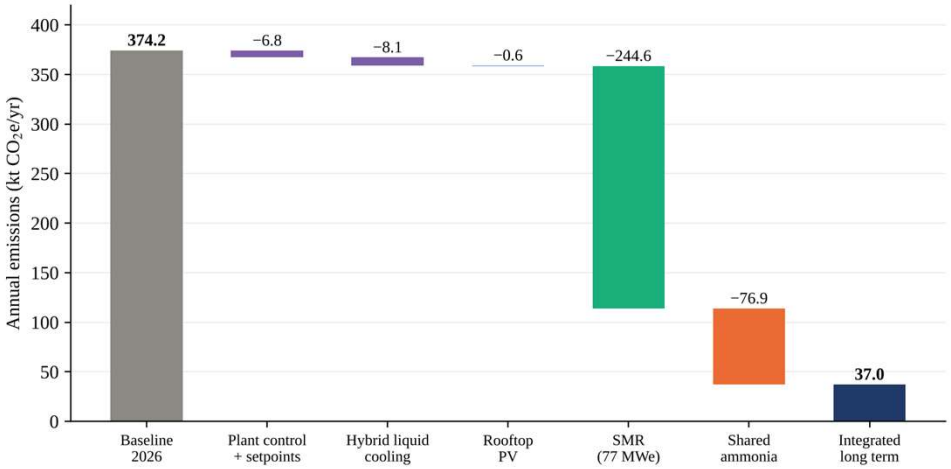


Figure 5.4. Annual emissions of the reference data centre from the 2026 baseline to the integrated long-term configuration, by measure

90% reduced emissions based on our impleemtnation. The long-term configuration takes emissions from 374.2 kt to 37.0 kt. That is 337.1 kt, which is 90.1% of the baseline (Table 5.3).

Measure	Hazard	Engineering control
PIML-MPC plant control	Setpoints outside the thermal band; model drift; cyber intrusion	Physics constraints, hard limits in the BMS and automatic fallback to fixed setpoints; segregated control network with read-only optimiser paths
Warmer halls (SS 697)	Server faults and higher IT fan power	Operate at the measured total-power optimum; agree the ASHRAE envelope in tenant contracts [12]
Hybrid liquid cooling (SS 726)	Coolant leaks, corrosion, biofilm, floor loading	Leak detection and isolation, fluid-quality monitoring, compatible piping and structural checks [16]
Cooling towers	Legionella in warm recirculating water	Continuous biocide treatment, routine sampling and drift elimination
Chillers	Fugitive leakage of high-GWP refrigerant (Scope 1)	Leak detection, annual audits and low-GWP refrigerant at chiller renewal
UPS batteries	Thermal runaway and fire	Fire-rated separation, cell-level monitoring and suppression
Ammonia GTCC and fuel handling (Jurong Island)	Ammonia toxicity and corrosivity; leaks; flame instability; NO_x, N_2O and ammonia slip	Gas detection with automatic isolation, ventilation, secondary containment and separation distances; staged low-NO_x combustion with SCR and continuous emissions monitoring [58], [34]; emergency plans coordinated with neighbouring plants
SMR (long term)	Nuclear safety, security and waste	National regulatory framework under the IAEA Milestones Approach, prepared through NEA safety studies and the 2027 INIR mission [61], [19]; the operator contracts for output only

Safety, Reliability, Implementaiton

Stakeholders

Industry

PSA

Keppel

STTelemedia

entres

EQUINIX

Policy and Regulatory

ENERGY MARKET AUTHORITY
Our Clean Energy Future

jtc

METI MINISTRY OF ENERGY, TRADE AND INDUSTRY SINGAPORE

Research Institutes

NUS National University of Singapore

Agency for Science, Technology and Research SINGAPORE

SNRSI
Singapore Nuclear Research and Safety Institute
16 Prince George's Park, Singapore 118415

REFLECTION

Reflection and Key Learning Points

What we learned

Net zero for data centres is a multidimensional challenge:

- **Energy efficiency**
- **Renewable integration**
- **Low-carbon power imports**
- **Policy and regulatory frameworks**

Key insights

- Efficiency improvements alone cannot achieve net zero if AI demand keeps growing rapidly.
- Reliability, affordability and sustainability must be balanced at the same time.
- Singapore's land constraints make regional energy partnerships essential.

Team reflection

We developed a stronger understanding of:

- Data-centre energy systems
- AI energy impacts
- Energy transition strategies
- Sustainability trade-offs

We saw why technical, economic and policy perspectives must be integrated when evaluating decarbonisation pathways.

Final reflection

Singapore's strategy is among the world's most comprehensive. Future success depends on managing growing AI demand while accelerating access to reliable low-carbon energy.